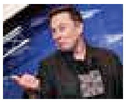

Elon Musk sued
Twitter shareholders issue lawsuit against Elon Musk for allegedly manipulating the company's stocks. **https://dgit.in/jun22-01**

OpenSea redesign NFT store
OpenSea has redesigned its website and profile to make it more easy to buy NFTs amid drop in NFT sales. **https://dgit.in/jun22-02**

AI HISTORIANS

Unravelling inscriptions has a new approach.

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Epigraphy, the study of inscriptions, is done to figure out how civilizations in out history communicated, thought, and how their society functioned. With time, these inscribed texts get damaged due to the harsh environmental conditions surrounding them or other natural causes. In such cases, it becomes difficult for historians to translate or understand what has been written on these inscriptions. This is where DeepMind comes in, with their new AI model, Ithaca. It is the first deep neural network which focuses on restoring lost and damaged texts, helping in deciphering, dating and locating these ancient inscriptions.

Ithaca is trained on the data of around 178,551 Greek texts, with the location and date labelled with the metadata. Similar to other machine learning systems, the information encoded by Ithaca is turned into mathematical models and identical patterns that have been noticed to decipher information about the inscriptions.

Ithaca by itself achieves up to 62% accuracy when restoring the inscriptions. When done by historians, it was at 25%. When historians used Ithaca, the accuracy peaked from 25% to 72%. Ithaca can date them within a range of up to 30 years. The table below shows how collaborative work between historians and AI can help uncover history in a way we have never seen before.

These numbers are impressive, but we need to remember that Ithaca is not capable of operating independently without human involvement. DeepMind has stated that is meant only for forensic aid and nothing more than that.

	Restoration			Region		Date
Method	CER↓	Top-1↑	Top-20↑	Top-1↑	Top-3↑	Years↓
Ancient Historian & Ithaca	18.3%	71.7%				
Ithaca	18.3%	61.8%	78.3%	70.8%	82.1%	29.3
Pythia	47.0%	32.6%	53.9%			
Ancient Historian	59.6%	25.3%				
Onomastics				21.2%	26.5%	144.4

Source: DeepMind

HOW DOES ITHACA WORK?

Ithaca is developed in a way to accommodated three important requirements. Firstly the AI must contextualise the texts, in this case, use words to represent the input. A problem that arises in this situation is that parts of the word might not be legible, and could've been lost years ago. Here, the text is input as a character and word together, and representing any damaged words with a symbol.

As previously mentioned, Ithaca is based on neural network architecture. So the AI uses an attention mechanism to assess the importance of different parts of the input. The attention mechanism finds the position of the input text and links the input character with the input word. Ithaca also consists of a transformer (neural network architecture) blocks to help figure out the length which equals the input characters. The output of each